prototype_verfahrweg-Copy1

October 22, 2020

1 Prototyp

1.1 Importing gcode in csv-file

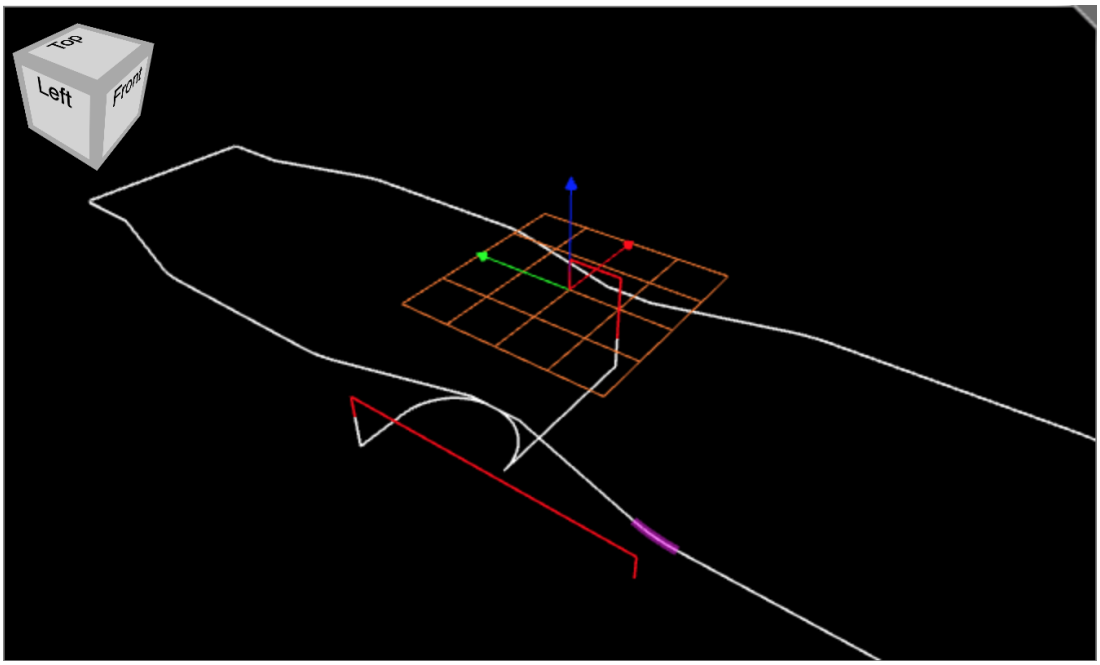
[1]:
```
#import data from csv file for further processing
import pandas as pd
import numpy as np

#small dataset
#td_filepath = './dataset4_sorted.csv'
#extended dataset
td_filepath = './dataset4 extended/dataset4_sorted_extended.csv'
td = pd.read_csv(td_filepath, index_col=['index'])
td.rename(columns={'Energy_consumed_kWs':'Energy_consumed_Ws'}, inplace=True)
td.head()


#td.tail()
#td.columns
```

[1]:

	G	X	Y	Z	I	J	F	Energy_consumed_Ws
index								
N288	0	0.0	0.0	0.0	NaN	NaN	0	0
N290	0	0.0	0.0	3.0	NaN	NaN	0	17
N292	0	0.0	-5.0	3.0	NaN	NaN	0	43
N294	0	0.0	-5.0	-3.0	NaN	NaN	0	16
N296	1	0.0	-5.0	-6.0	NaN	NaN	1000	25



1.2 Make information in data visible for ML-model

[2]:
```
# inspect data here

td.describe()
```

[2]:

	G	X	Y	Z	I	J	\
count	97.000000	97.000000	97.000000	97.000000	42.000000	41.000000	
mean	1.391753	-3.380144	-1.503289	-4.237113	-0.193643	0.281146	
std	0.757693	12.131840	34.396031	1.977863	9.589312	2.977698	
min	0.000000	-19.907000	-62.907000	-6.000000	-14.300000	-7.300000	
25%	1.000000	-13.300000	-18.920000	-6.000000	-7.941000	-1.889000	
50%	1.000000	-9.517000	-2.524000	-3.000000	0.000000	0.000000	
75%	2.000000	10.406000	15.478000	-3.000000	7.941000	2.148000	
max	3.000000	14.300000	62.907000	3.000000	14.300000	7.300000	

	F	Energy_consumed_Ws
count	97.000000	97.000000
mean	927.835052	157.938144
std	260.105054	211.185040
min	0.000000	0.000000
25%	1000.000000	17.000000
50%	1000.000000	41.000000
75%	1000.000000	306.000000
max	1000.000000	776.000000

[3]:
```
#processing data in table to add features
from math import sqrt


#calculating travelled distance
distances = []
distances.append(0)
for i in range(1, len(td.index)):
    distances.append( sqrt(pow((td.X.iloc[i]-td.X.iloc[i-1]), 2) + pow((td.Y.
 ↪iloc[i]-td.Y.iloc[i-1]), 2))) #+ pow((td.Z.iloc[i]-td.Z.iloc[i-1]), 2)
try:
    #td.insert(9, 'delta_Z', delta_z)
    td['traveled_distance'] = distances
except:
    pass

#create delta_z for differentiated calculations
delta_z = []
delta_z.append(0)
for i in range(1, len(td.index)):
    delta_z.append(td.Z.iloc[i] - td.Z.iloc[i-1])
try:
    #td.insert(9, 'delta_Z', delta_z)
    td['delta_z'] = delta_z
except:
    pass

#calculating delta F
delta_F = []
delta_F.append(0)
for i in range(1, len(td.index)):
    delta_F.append( td.F.iloc[i] - td.F.iloc[i-1] )
try:
    #td.insert(9, 'delta_Z', delta_z)
    td['delta_F'] = delta_F
except:
    pass

td.head(7)
#td.tail()
```

[3]:

	G	X	Y	Z	I	J	F	Energy_consumed_Ws	\
index									
N288	0	0.000	0.0	0.0	NaN	NaN	0	0	
N290	0	0.000	0.0	3.0	NaN	NaN	0	17	
N292	0	0.000	-5.0	3.0	NaN	NaN	0	43	
N294	0	0.000	-5.0	-3.0	NaN	NaN	0	16	

```
N296   1   0.000 -5.0 -6.0  NaN  NaN  1000                  25
N298   1 -15.406 -5.0 -6.0  NaN  NaN  1000                  50
N300   3 -10.406  0.0 -6.0  0.0  5.0  1000                 156

       traveled_distance  delta_z  delta_F
index
N288            0.000000      0.0        0
N290            0.000000      3.0        0
N292            5.000000      0.0        0
N294            0.000000     -6.0        0
N296            0.000000     -3.0     1000
N298           15.406000      0.0        0
N300            7.071068      0.0        0
```

[4]:
```
#safe csv for external inspection
td.to_csv (r'./dataset4_export_from_pandas.csv', index = False, header=True)
```

1.3 Visualize Data and show dependencies

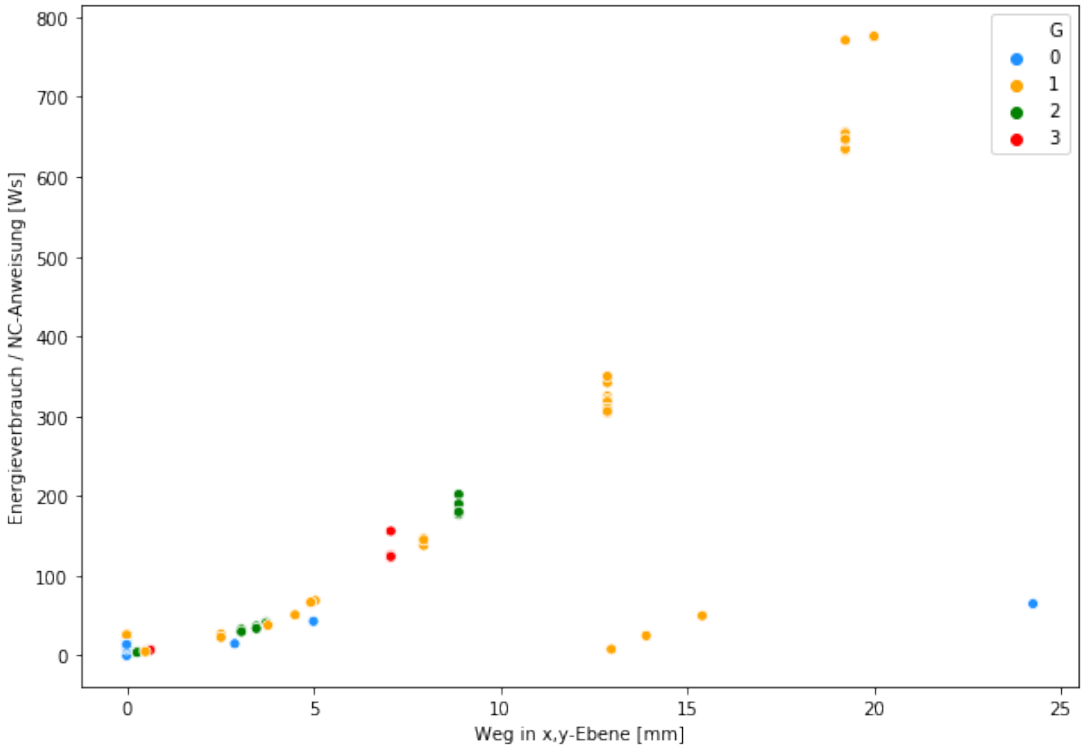
[5]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Examine the data by sight

#td.plot(figsize=(30,10))

plt.figure(figsize = (10, 7))
sns.scatterplot(x = td['traveled_distance'],
                y = td['Energy_consumed_Ws'],
                hue = td['G'],
                palette=['dodgerblue','orange', 'green','red'],)
plt.xlabel("Weg in x,y-Ebene [mm]")
plt.ylabel('Energieverbrauch / NC-Anweisung [Ws]')
```

[5]:
```
Text(0, 0.5, 'Energieverbrauch / NC-Anweisung [Ws]')
```

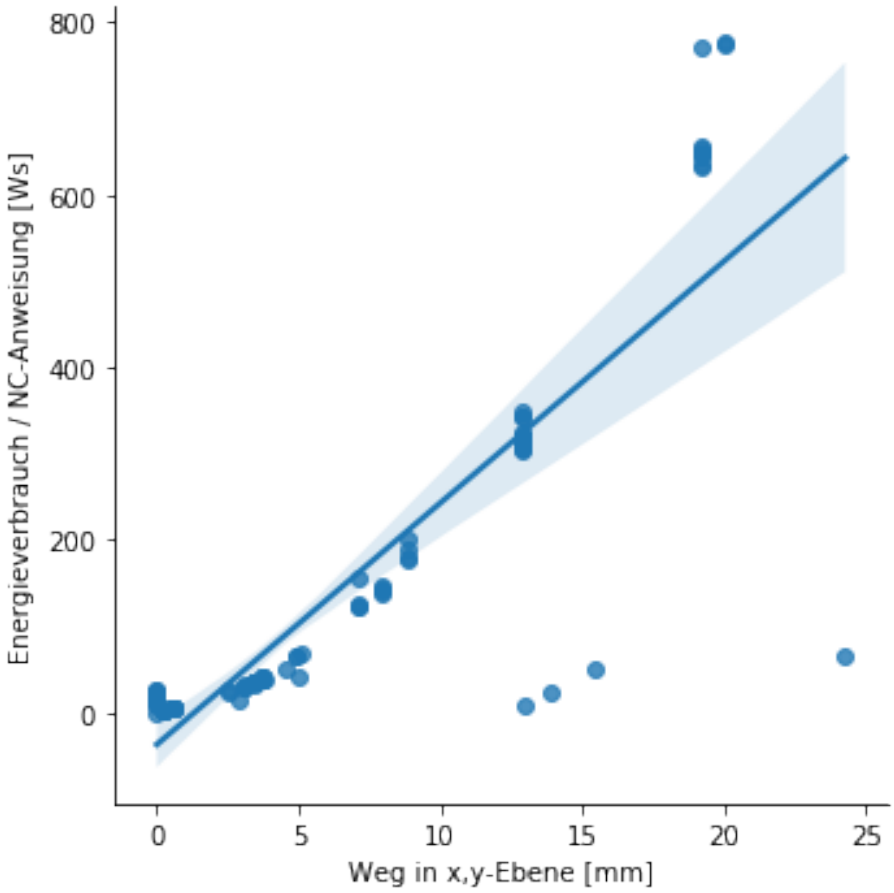


[6]:
```
plt.figure(figsize = (10, 7))

sns.lmplot(x = 'traveled_distance',
           y = 'Energy_consumed_Ws',
           #hue = 'G',
          data = td)
plt.xlabel("Weg in x,y-Ebene [mm]")
plt.ylabel('Energieverbrauch / NC-Anweisung [Ws]')
```

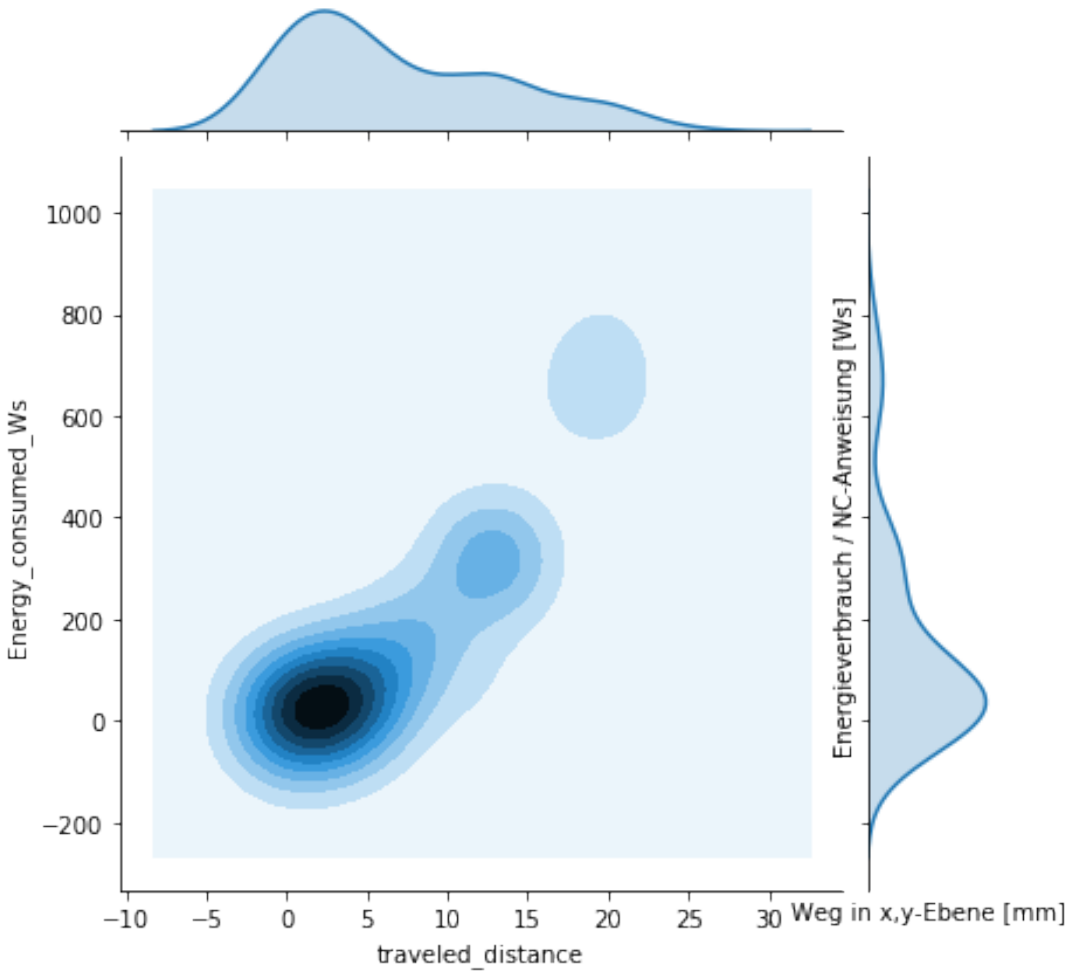
[6]: Text(3.799999999999997, 0.5, 'Energieverbrauch / NC-Anweisung [Ws]')

<Figure size 720x504 with 0 Axes>



[7]:
```
sns.jointplot(x = td['traveled_distance'],
              y = td['Energy_consumed_Ws'],
              kind="kde")
plt.xlabel("Weg in x,y-Ebene [mm]")
plt.ylabel('Energieverbrauch / NC-Anweisung [Ws]')
# Fazit: Problem bei kleinen und großen Wegen
```

[7]: Text(336.9714285714286, 0.5, 'Energieverbrauch / NC-Anweisung [Ws]')



1.4 Split dataset and train model

[8]:
```
#split dataset into train and test dataset

from sklearn.model_selection import train_test_split


td_cleaned = td.iloc[1:,]
y = td_cleaned.Energy_consumed_Ws
td_features = ['G', 'traveled_distance', 'delta_z'] #  , 'delta_F', if included␣
↪results get worse. F is is feed speed and get's ramped up every time anyway
X = td_cleaned[td_features]
```